

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL

Tools/Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, Docker, FastAPI, AWS SageMaker, Kubernetes, Git, Airflow

Competence: Machine Learning, Deep Learning, MLOps, Computer Vision, Natural Language Processing (LLMs), Data Engineering, Data Science, Perception, Segmentation, Trajectory Prediction, Reinforcement Learning, Agile Development, Continuous Integration

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** for **metrological applications**, enhancing **model performance** and **training efficiency**.
- Explored **novel architectures** and physics-informed hybrid loss functions to significantly improve **model accuracy** and **stability**.
- Collaborated with PhD researchers on a **publication** presented at the FLOMEKO conference, contributing to scientific advancement.
- Developed interactive dashboards for real-time **data visualization** and comprehensive **model performance monitoring**.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Achieved 94% **segmentation accuracy** for vitiligo detection and boosted inference speed by 5x using advanced **AI algorithms**; deployed solutions with Docker and FastAPI.
- Developed a **real-time** tennis foul **detection system** leveraging **TensorRT**, reducing compute cost and processing time by 6x for **edge processing**.
- Managed robust **MLOps pipelines** on **AWS SageMaker**, effectively addressing **YOLO model data drift** and maintaining **Kubernetes infrastructure**.
- Engineered **Proof-of-Concepts (POCs)** using **YOLO**, **few-shot learning**, and **CNN** architectures for impactful client demonstrations.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Contributed to full-stack development of an AI-based services website using Flask, HTML, and CSS.
- Developed an **AI chatbot application** by integrating the **OpenAI API** and relevant **LLM tools/frameworks**.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built and optimized **web scrapers** and **ETL pipelines** for **structured and unstructured data** ingestion.
- Automated **data ingestion** and **cleaning workflows**, significantly improving data readiness for client **analytics use cases**.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a **weighted-probability ensemble** integrating DPFunc and DeepGOPlus, utilizing AlphaFold for robust **feature extraction** and **taxonomic bias mitigation**.
- Increased **model accuracy** by 4% through innovative **custom feature engineering** across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a **time series prediction** model for **real-time Bitcoin price forecasting**, achieving high accuracy using modern ML techniques.
- Engineered **data pipelines** to ingest and process cryptocurrency market data for analysis and prediction.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to build the world's first **AI-driven** self-playing Chitralli Sitar, integrating **signal processing** and **music generation LLMs**.
- Project gained significant recognition, featured on Coke Studio in a music video by Hassan Raheem.

Hugging Face

Open-Source

Contributor

- **Co-authored** the **Zero-shot Image Classification** technical **documentation** and implementation page on the Hugging Face Hub, facilitating broader community usage.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specializing in **Deep Learning**, **Computer Vision**, and **Autonomous Systems**.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Awarded Dean's List of Honor for academic excellence.
- Focused on **Machine Learning fundamentals**, **data structures**, and **algorithms**.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python